Journal of Open Aviation Science

JOAS

ARTICLE (***Pre-print***)

Predicting Aircraft Taxi-Out Times from Imperfect Clocks and Open Surveillance Data

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Abstract

Taxi-out time is recorded by airports as the difference between take-off and an off-block time that is itself imperfect: some recorded off-block times are copies of planned times or are shifted by one day, and under squared error these few records dominate the total error. This paper addresses the post-operational prediction of taxi-out times at ten large European airports for four months of 2026 in the EUROCONTROL Performance Review Commission Data Challenge 2026, with the 2025 movement records as the only labels. The problem is formulated as the estimation of an unobserved moment from several imperfect clocks. Three properties of the data-generating process are used as model structure: the aircraft's own ground track, extracted from public ADS-B archives and anchored at the assigned stand, provides an independent clock of the pushback; the rule by which NM flights were joined to departures bounds every label; and a daily pre-departure delay published by EUROCONTROL equals, on the 2025 labels, a known function of the labels aggregated per airport and day, which constrains each airport-day's predictions jointly. Gradient-boosted residual learners, two mixed-target members, neighbourhood corrections, a Bayesian reconciliation with the published total and data-derived bounds are combined, and every component is admitted by one selection rule: it must reduce the error in each 2025 month that mirrors a test month, its weight is verified in nested cross-validation, and gains concentrated in the hundred most-improved records are rejected. On the 2025 out-of-fold validation the pipeline attains an RMSE of 246.9 s on the four analogue months (273.6 s for the base learners), and 234.55 s on the public January/July 2026 leaderboard; the ground track and the daily reconciliation contribute most. A residual-learnability test indicates that the available features are exhausted, so further improvement requires new observations of the off-block moment. The code is reproducible and period-independent.

Keywords: taxi-out time; ADS-B; gradient boosting; model selection; airport operations

Abbreviations: ADS-B: Automatic Dependent Surveillance–Broadcast, AOBT: actual off-block time, LOBT: last off-block time, NM: Network Manager, RMSE: root-mean-square error

1. Introduction

Taxi-out time is a principal determinant of fuel consumption, emissions and the predictability of departure sequences at congested airports [1, 2]. The Data Challenge 2026 of the EUROCONTROL Performance Review Commission (PRC) [3] poses its post-operational prediction for all departures of ten large European airports, referred to below by their ICAO location indicators: Frankfurt (EDDF), Munich (EDDM), London Heathrow (EGLL), Amsterdam Schiphol (EHAM), Barcelona (LEBL), Madrid Barajas (LEMD), Paris Charles de Gaulle (LFPG), Rome Fiumicino (LIRF), Zurich (LSZH) and Istanbul (LTFM). The training labels cover the twelve months of 2025; the final evaluation covers January, February, June and July 2026 (670,790 departures) and is blind. Performance is measured by the RMSE in seconds over all scored records.

Three characteristics of the task determine the approach taken here. First, the take-off time of every departure is known, whereas the off-block time is not; several clocks nevertheless carry information

about it, namely the schedule, the planned off-block times filed with the NM (estimated, last and initial off-block times) and the NM estimate of the actual off-block time. Second, the labels are produced by airport systems that contain recording defects, and these records remain in the scored set, where their squared errors dominate. Third, the evaluation months follow the training year, so that model selection must favour what persists over time rather than what fits a single period.

The contributions of this paper are as follows.

- An open-data pipeline that extracts, from the public adsb.lol daily archives [4], the time at which each departing aircraft began to move and associates it with the departure (Section 4.4). It yields the largest single reduction in error (13.7 s).
- Three properties of the data-generating process that are used as model structure rather than as features: a hard bound on the label implied by the NM join rule and airport-specific maxima, both without exception on 2025, and a near-identity (correlation 0.9996) between a published daily delay total and the labels (Section 3).
- A Bayesian reconciliation of the predictions of each airport-day with that published total, including an exact posterior for records with two discrete recording modes (Section 4.6), which reduces the error by 8.7 s.
- A single selection rule for all components, with a concentration criterion against gains carried by a few records, together with an audit of its decisions and a set of measured negative results (Sections 5 and 7).
- A reproducible, period-independent implementation in which every decision is stored and the components can be reused separately (Section 9).

The remainder of the paper is organised as follows. Section 2 describes the data, Section 3 analyses the structure of the error, Section 4 presents the method, Section 5 the validation and selection procedure, Section 6 the results and Section 7 the rejected alternatives; Section 8 discusses the findings and Section 10 concludes.

2. Data

2.1 Challenge data

The organisers provide movement records for the ten airports. For each departure the records contain the take-off time, the scheduled time, stand, runway, aircraft type and operator and, where an NM flight could be associated, the NM fields: the estimated off-block time at first filing, the LOBT, the initial off-block time, the NM AOBT and the flight type. The label is $y = t_{\mathrm{to}} - t_{\mathrm{ob}}$, the difference in seconds between take-off and the recorded off-block time. The 2025 training set comprises 2,085,047 departures. Arrival records are also provided and retain their block and taxi-in times in the scored file, which the data description permits to use; for departures in the scored file the block time and the label are withheld. The delay to take-off, $d = t_{\mathrm{to}} - t_{\mathrm{sched}}$, is known for every record.

2.2 External data

Three external sources are used (Table 1); all are openly accessible and none contains a label.

3. Structure of the error

3.1 Concentration and recording regimes

The squared error of the final model on the 2025 out-of-fold predictions is strongly concentrated: the 100 largest errors account for 20 % of the total, the 1,000 largest for 32 % and the 10,000 largest

Table 1. External data sources.

Source	Use	Licence
adsb.lol `globe_history` daily archives [4]	ground tracks of departing aircraft (485 daily archives of about 3 GB each)	Open Database License 1.0
Iowa Environmental Mesonet archive of aerodrome weather reports (METAR) from automated surface observing stations [5]	hourly weather at each airport	open access
EUROCONTROL all pre-departure delay [6]	daily delay total per airport (Section 4.6)	attribution, non-commercial

for 50 % (Figure 7a). Table 2 decomposes the squared error according to the relation between the label and the available clocks. Records without a recognisable defect account for 81 % of the squared error; 147 records, about half of them at LIRF, whose label is shifted by approximately one day or is otherwise extreme, account for 14 %; and 2,016 records whose recorded off-block time coincides with the schedule to the second account for a further 2.5 %. The defective records can be identified with high accuracy (a classifier separates the corrupt LIRF records with an area under the curve of 0.99), but the magnitude of their error cannot be predicted from any available field. This observation motivates the concentration criterion of Section 5.

Table 2. Share of the records and of the 2025 out-of-fold squared error ("Error") by recording regime (twelve months, excluding the five corrupt early-day records). A record is classed as a copy of a clock when its recorded off-block time equals that clock within one second.

Regime	Records	Records (%)	Error (%)	RMSE (s)
no recognisable defect	2,011,282	96.5	81.5	191
shifted by about one day, or extreme	147	0.01	13.9	9,251
copy of the schedule	2,016	0.10	2.5	1,053
copy of the LOBT	27,019	1.30	1.3	211
equal to the NM AOBT	36,700	1.76	0.6	118
copy of the first estimated off-block time	7,744	0.37	0.2	142

3.2 Properties of the data-generating process

Join window. Let $\Delta_{\mathrm{L}} = t_{\mathrm{to}} - t_{\mathrm{LOBT}}$. All 2,062,577 departures of 2025 that carry an NM flight satisfy $|t_{\mathrm{ob}} - t_{\mathrm{LOBT}}| \le 3{,}606$ s, that is, $\Delta_{\mathrm{L}} - 3606 \le y \le \Delta_{\mathrm{L}} + 3606$. This is interpreted as the rule by which NM flights were joined to departures; it bounds every label for which a LOBT exists.

Airport maxima. No taxi-out time at LEMD or LEBL exceeded 3,604 s in 2025.

Published daily total. EUROCONTROL publishes, for each airport a and day t, the total pre-departure delay in minutes (DLY_ALL_PRE_2). On the 2025 labels,

$$K_{a,t} \equiv 60\,\mathrm{DLY_ALL_PRE_2}_{a,t} \approx \sum_{i \in (a,t)} \max(0,\ d_i - y_i), \tag{1}$$

with a correlation of 0.9996 and a median ratio of 1.0001 over 3,650 airport-days (Figure 5a). The published total is therefore the sum of the off-block delays relative to the schedule. Because every d_i is known, it constrains the unknown labels of an airport-day jointly.

Seasonality of long NM clock gaps. NM AOBT values more than one hour before take-off are predominantly recording errors in summer (July 2025: median prediction 993 s against a median label of 958 s) but mostly genuine in winter, when de-icing takes place after the aircraft has left the stand (January 2025: 3,066 s against 3,237 s). The model can learn this distinction from calendar and weather features, and such clocks are therefore retained rather than masked.

4. Methods

Figure 1 summarises the pipeline. All learners are gradient-boosted decision trees, implemented with LightGBM [7] or CatBoost [8], and successive stages are trained on out-of-fold predictions of the preceding stage [9].

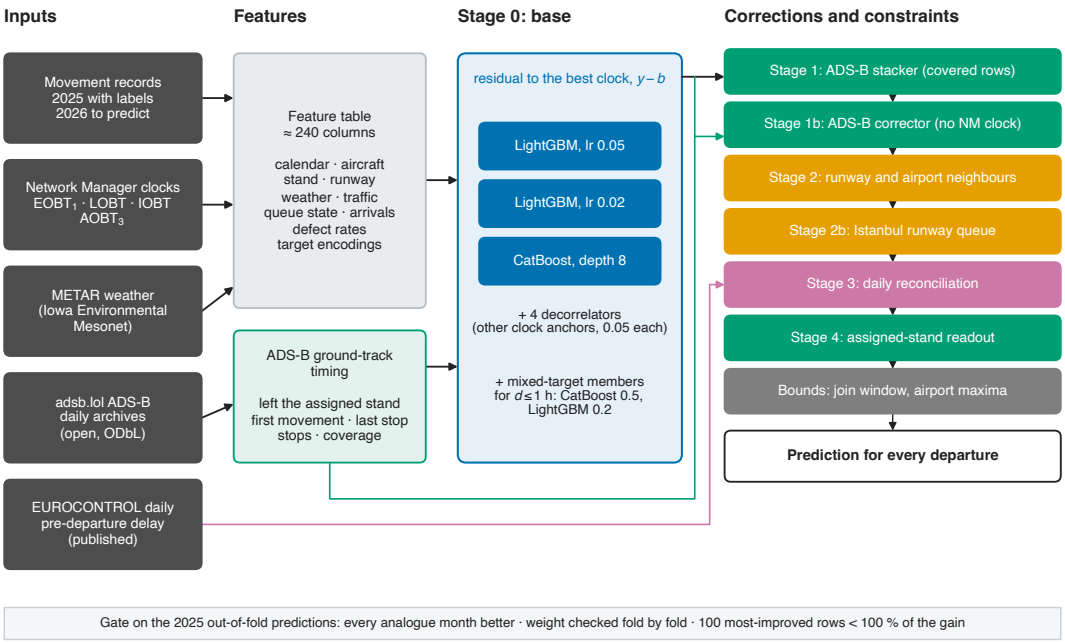


Figure 1. Pipeline overview. Stage 0 predicts the residual with respect to the best off-block clock; subsequent stages use evidence available only for subsets of records (lanes); stage 3 reconciles each airport-day with the published total; stage 4 moves covered records towards the reading of the assigned-stand clock where the two agree; data-derived bounds are applied last. Every component is admitted by the selection rule of Section 5.

4.1 Target definition

For each record let b denote the time from the most reliable plausible off-block clock to take-off: $b = t_{\text{to}} - t_{\text{AOBT}}$ if this value lies in $[-2\,\text{h}, 48\,\text{h}]$, otherwise the corresponding value for the LOBT and then for the first estimated off-block time, and 1,000 s if no clock is plausible; b is never negative. Departures from LIRF without an initial NM time use $b = d$. The learners predict the residual $r = y - b$, and the prediction is $\hat{y} = b + \hat{r}$ (Figure 2). The NM clock explains most of the variance of the label; the residual describes how far the recorded off-block time deviates from it because of

queues, stand positions, de-icing and recording defects, and it is comparable across airports whose taxi times differ by several minutes. All records, including defective ones, are retained in training (the correction stages leave out only the five corrupt day-early records): every variant that removed, down-weighted or capped them improved the accuracy on flights without defects but increased the overall RMSE, because such records are part of the scored set.

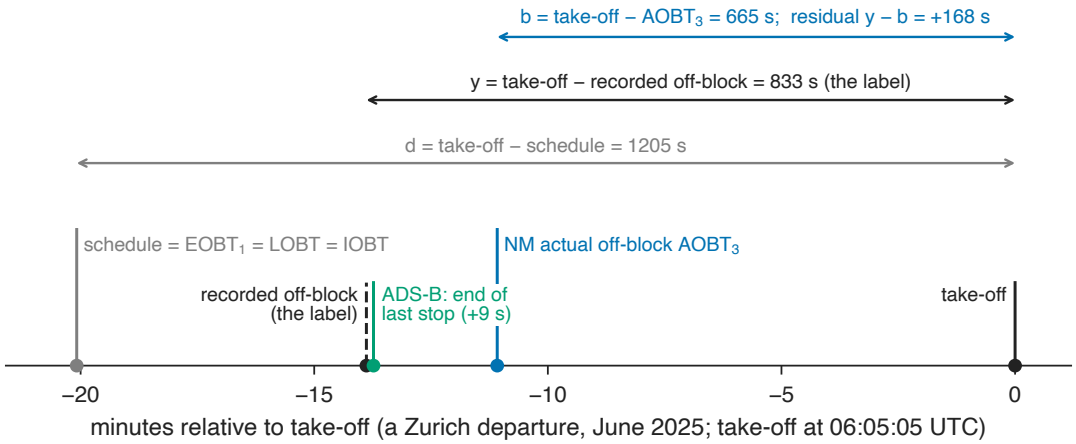


Figure 2. Clocks of one departure from LSZH (June 2025). The planned clocks coincide with the schedule; the NM AOBT lies 168 s after the recorded off-block time, whereas the end of the last stop in the aircraft's ADS-B track lies 9 s after it. The label y extends from the recorded off-block time to take-off, d from the schedule to take-off, and the learners predict the residual with respect to the anchor b.

4.2 Lanes defined by the available evidence

The error of a record depends primarily on the evidence available for it. Records are therefore partitioned by two properties that are also known for the evaluation period (Table 3 and Figure 3): the presence of an NM clock and the reception of the aircraft's own ground track. Partitioning by evidence rather than by airport allows a single stage to serve every airport at which its evidence exists.

Table 3. Lanes, their number of records in the 2026 evaluation file (column 2026) and their 2025 out-of-fold RMSE in seconds per analogue month. "Own track" denotes a usable ADS-B ground track of the departing aircraft.

Lane	Definition	2026	Jan	Feb	Jun	Jul
covered	NM clock, own track	454,526	118.5	101.7	167.1	206.5
uncovered	NM clock, no track	204,189	214.6	245.5	207.6	215.6
unmatched, other	no NM clock, not LIRF	11,393	2,467	1,083	392	537
unmatched, LIRF	no NM clock, LIRF	682	3,968	1,798	6,756	3,407

4.3 Stage 0: base learners and additional members

Three learners are trained on approximately 240 features that do not depend on ADS-B (calendar, aircraft, stand and runway, METAR weather, traffic and queue state around the departure, arrivals, recording-defect rates of the day and out-of-fold target encodings; Table 8 in the appendix lists the feature groups): two LightGBM models with learning rates of 0.05 and 0.02 and a CatBoost model of depth 8. Missing values are not imputed: both libraries route them natively in each split, which

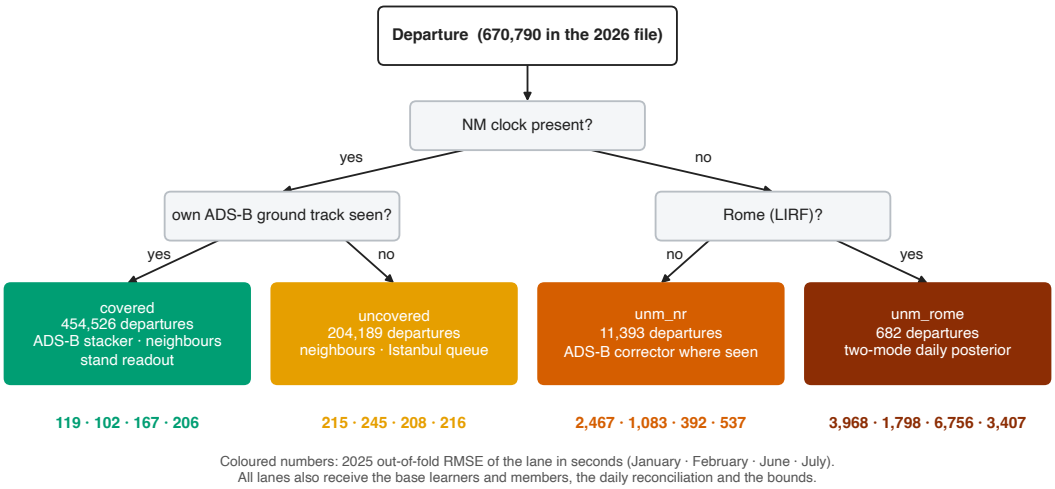


Figure 3. Assignment of a departure to a lane, with the number of records in the 2026 evaluation file and the stages applied in each lane.

preserves the information that a clock or an observation is absent. The settings of every model are listed in Table 7. The base prediction is 0.45 times the mean of the two LightGBM models plus 0.55 times the CatBoost model, the optimum over the twelve months (0.50 to 0.60 when each fold is left out).

Decorrelators. Four LightGBM models predict the residual with respect to other anchors (the LOBT, no anchor, the first estimated off-block time and the initial off-block time). Where an NM clock exists they enter with a weight of 0.05 each; they err differently when the NM clock itself is wrong.

Mixed-target members. A model with a different target constitutes a different model. For departures with $d \le 3{,}600$ s the member predicts $\min(y, 7{,}200)$ directly; for later departures it predicts the share of the delay spent after off-block,

$$\tilde{y} = \min\left(\max\left(\frac{y}{d}, 0\right), 1.5\right), \qquad w = \left(\frac{\min(d, 43{,}200)}{3{,}600}\right)^2, \qquad \hat{y} = \hat{\tilde{y}}\, d, \tag{2}$$

(Equation 2), with sample weights w because the squared error of a long-delay record, measured in seconds, scales with d^2. A CatBoost version (depth 10, 3,000 iterations, trained on graphics processors and averaged over eight seeds) is combined with the base in equal parts for departures with $d \le 1$ h, and a LightGBM version enters with a weight of 0.2 for the same departures. The admissible scopes (all records and departures with $d \le 1$ h, and for the LightGBM version also the uncovered records) were fixed before any result was observed; the variant applied to all records reduced the 2025 error further but failed the concentration criterion (Section 5).

4.4 Stage 1: the aircraft's own ground track

For every day of 2025 and of the four evaluation months, the adsb.lol archive is retrieved and the positions within ±0.15° of each airport are retained. These positions are associated with departures by callsign (flight numbers are translated into callsigns with an airline-designator table) and, failing

that, by aircraft address. Within the three hours before take-off the following quantities are extracted: the first sample, the first ground sample, the first sustained movement (ground speed above 5 kn), the end of the last stationary episode preceding it, the first fast sample (above 15 kn), the number of stops, a coverage code and the share of the airport-day's departures observed (Figure 4). These timings constitute an independent clock of the actual pushback. The raw positions are cleaned as follows: only samples inside the airport box and within the three-hour window are kept; ground samples are identified by the transponder's ground flag; a sample is stationary if its ground speed is at most 2 kn, and stationary episodes are summarised at several minimum durations between 30 and 900 s, which removes isolated speed spikes; a departure is labelled with a coverage code according to whether no sample, only airborne samples, moving ground samples or a stationary ground episode was observed, and days for which no archive could be retrieved receive a separate code. Coverage is good at seven airports, partial and unstable at LFPG and LIRF, and absent at LTFM; 65 to 72 % of the 2026 departures are covered.

Assigned-stand timing. The end of the last stop confuses the stand with taxiway holds, de-icing pads and runway queues. Since every departure record names its stand, the stand position is learned from the data. A first extraction pass records each departure's parking position, the median position of the stationary samples of the stop that ends at pushback. On 2025, a departure whose stop ended within 60 s of its recorded off-block time was parked at its stand; the median of these parking positions locates the stand, which is retained if it has at least three observations whose median distance to it is at most 30 m. This yields 749 stands with a median scatter of 3 m; positions learned separately from odd and from even months agree to 1.3 m, so the labels only select clean observations and do not leak into the features. Arriving aircraft locate further stands without any label, since an arrival stops at its stand: the median of the slow ground samples around the in-block time of 2025 arrivals agrees with the departure-located position to 2.6 m (median). At EDDF, EDDM, EHAM, LEBL and LSZH these positions add 180 stands (929 in total); at EGLL, LEMD and LIRF they are not used, because the departing aircraft is rarely received at its stand before pushback and the readings align with the recorded off-block in only 1 to 10 % of cases. A second pass reads, for every departure whose assigned stand is located, the last sample within 40 m of the stand before the aircraft is next seen more than 80 m away. This reading exists for about 14 % of the departures in both years; on those records its median error is 39 s, against 172 s for the end of the last stop and 128 s for the NM AOBT. The label rule is what makes a position a stand: the median of all parking stops at a stand, without it, would add some 60,000 departures, but on the 2025 labels their readings lie 200 to 380 s after the recorded off-block (0 to 10 % within 60 s, against 59 to 85 % for label-selected stands), because the stop most often observed is the hold after pushback. Records that no callsign or aircraft address links to a track are additionally linked to the unique unclaimed aircraft that lifts off within ±60 s of their take-off.

For covered records, a LightGBM stacker learns $y - \hat{y}_{\text{base}}$ from all features, the ADS-B timings relative to the NM AOBT and the base prediction, trained on out-of-fold base predictions. Departures without an NM clock outside LIRF whose track reaches take-off (40 % of them in 2025 and 50 % in 2026) receive an analogous correction (stage 1b), trained on these records with weight 1 and on covered records with weight 0.2.

4.5 Stage 2: neighbourhood corrections

Errors are clustered in time on the same runway: when the clock of one departure is late, those of its neighbours tend to be late as well. A small LightGBM model predicts the remaining error of a flight from neighbourhood aggregates that exclude the flight itself: the mean ADS-B correction of covered neighbours within ±5, 15 and 30 min on the same runway and within ±15, 30 and 60 min

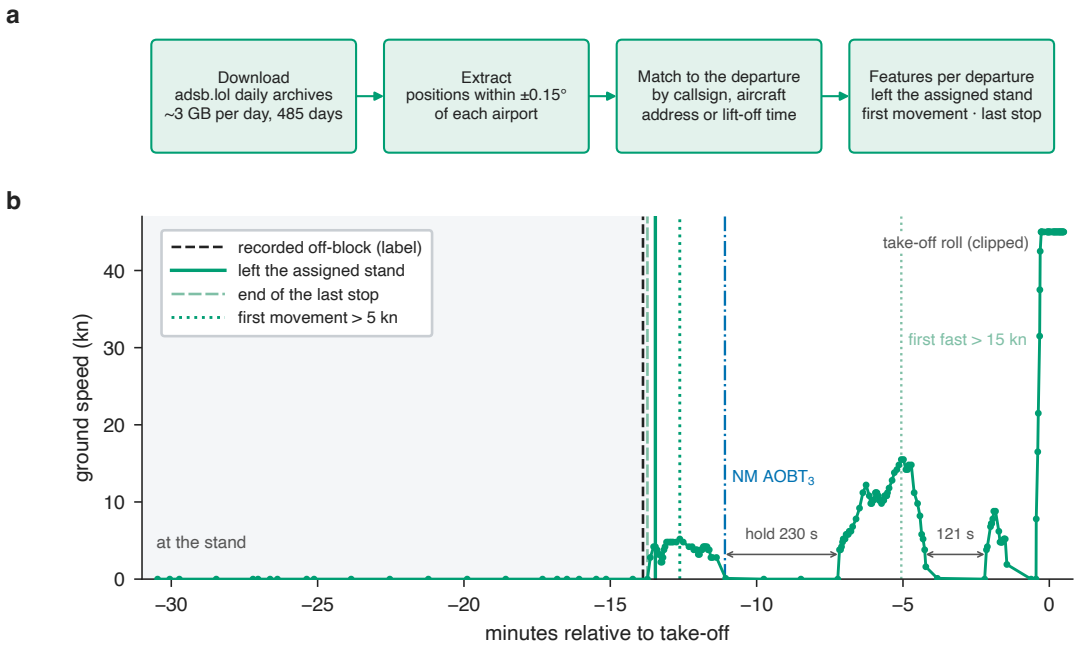


Figure 4. (a) Extraction of per-departure timing features from public ADS-B archives. (b) Ground speed of the departure shown in Figure 2, obtained from its adsb.lol track, with the extracted events, the recorded off-block time and the NM AOBT; two holding periods during taxiing are visible.

at the airport, and the mean stage-1 offset of matched neighbours. It is applied with a shrinkage factor of 0.75. At LTFM, where no ground coverage exists and the NM AOBT is predominantly a default value, the interval between that clock and take-off reflects the runway queue; stage 2b learns it from the neighbours' take-off delays relative to the LOBT and the schedule, expressed relative to the airport-day mean, and replaces stage 2 at this airport.

4.6 Stage 3: reconciliation with the published daily total

Equation 1 relates the published total of airport a on day t to the unknown labels. Let p_i denote the incoming predictions and call a record delayed if $p_i < d_i$. The discrepancy of the airport-day is

$$R_{a,t} = \left(K_{a,t} - \mu_a\right) - \sum_i \max(0,\ d_i - p_i) - \beta_a\, n_{a,t}, \tag{3}$$

where μ_a is the median offset between the published total and the label-implied sum on 2025, $n_{a,t}$ the number of delayed records and β_a the expected difference per delayed record that the convexity of $\max(0, \cdot)$ introduces between the true and the predicted sum. Each delayed record has a prior $y_i \sim \mathcal{N}(p_i, v_i)$, with v_i the 2025 out-of-fold error variance of its lane and delay band. The discrepancy of Equation 3 is distributed as in a Kalman update [10],

$$p_i \leftarrow p_i - \lambda\, \frac{v_i}{\sum_j v_j + s_a^2}\, R_{a,t}, \tag{4}$$

where s_a is the robust spread of the published total around the label-implied sum and $\lambda = 0.75$ in Equation 4 is selected by the rule of Section 5.

LIRF departures without an NM clock and with $d > 1$ h were recorded either at the schedule ($y = d$, no contribution to the total) or after a genuine hold of approximately $\tau = 1{,}020$ s ($y \approx \tau$, contribution

$d-\tau$). With the prior probability π_i of the first mode implied by $p_i = \pi_i d_i + (1-\pi_i)\tau$, the exact posterior over the 2^m mode combinations s of the day ($m \le 12$) is evaluated with the weights of Equation 5,

$$\log w(s) = \sum_i \log P(s_i) + \log t_\nu\left(\frac{R - \sum_i c_i(s_i)}{\sqrt{\phi\, S}}\right), \tag{5}$$

a Student-t likelihood with $\nu = 3$ and a variance inflated by $\phi = 4$, so that a record is moved only when the daily total is decisive; S denotes the variance of the remaining records. The posterior mean of each record replaces its prediction. All parameters are estimated on 2025 and validated by re-estimation without the held-out fold; for the 2026 file, the ±12 s take-off perturbation of the evaluation data is added to the noise. This layer is the only step that uses information about 2026 beyond the features of a record, and that information is a published aggregate rather than a label.

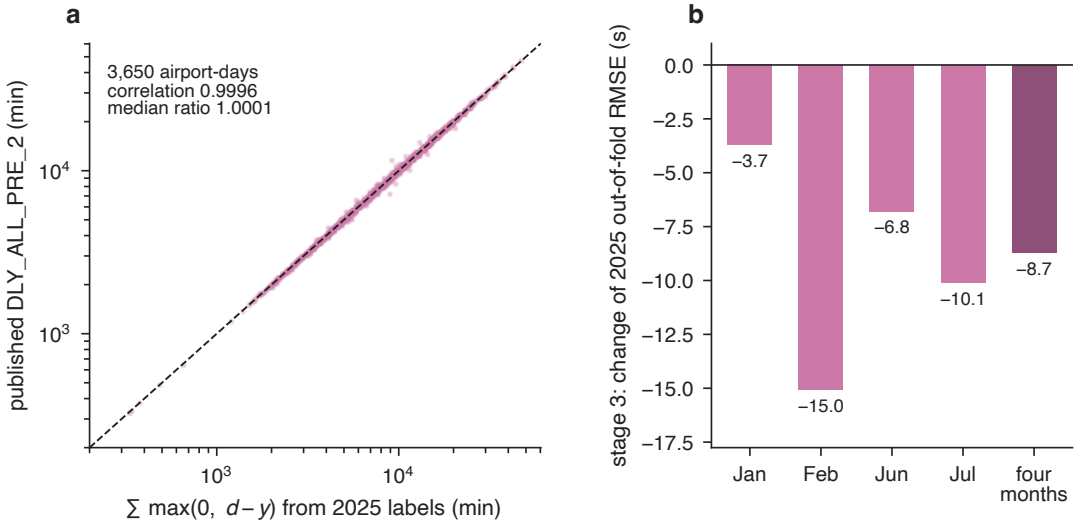


Figure 5. (a) Published daily pre-departure delay against $\sum \max(0, d - y)$ computed from the 2025 labels, one point per airport and day (Equation 1). (b) Change of the 2025 out-of-fold RMSE produced by the reconciliation in each analogue month.

4.7 Stage 4: readout of the assigned-stand clock

For a covered record with an assigned-stand leave time L (relative to take-off), $r = -L + o_a$ reads the taxi-out time directly, where the offset o_a is the 2025 median of $y + L$ at airport a. On the 2025 records with this reading, its median absolute error at EDDF, EDDM, LEBL and LSZH is 10 to 23 s, against 19 to 32 s for the prediction $\hat{y}$ of the preceding stages: the stacker uses the clock but blurs its precision. The readout fails, however, whenever the last sample at the stand is not the pushback. The final model step therefore moves $\hat{y}$ towards r with a weight that depends only on their agreement, $\hat{y} \leftarrow \hat{y} + w_b\,(r - \hat{y})$, where b is the bin of $|r - \hat{y}|$ with edges 15, 30, 60, 120, 300 and 600 s and w_b is the least-squares coefficient within the bin, clipped to $[0, 1]$; records with $|r - \hat{y}| \ge 600$ s are left unchanged. The fitted weights are 0.62, 0.63, 0.60, 0.42, 0.04 and 0.07, and differ by at most 0.03 when refitted without any one of the six folds. The step changes about 13 % of the predictions, by about 6.5 s in root mean square (6.4 s in 2025, 6.6 s on the evaluation file).

4.8 Bounds

The final step applies the join window, the airport maxima and one consequence of the definitions: for records with a LOBT, $\max(60, \Delta_L - 3606 - \delta) \le \hat{y} \le \Delta_L + 3606 + \delta$, with $\delta = 25$ s on the evaluation file

to account for its take-off perturbation; $\hat{y} \le 3{,}604$ s at LEMD and LEBL; and $\hat{y} \le d$ when $d > 7{,}200$ s. The bounds encode properties of the data and alter only predictions that violate them.

5. Validation and model selection

5.1 Protocol

The twelve months of 2025 are divided into six two-month folds, $\{1, 7\}$, $\{2, 8\}$, $\{3, 9\}$, $\{4, 10\}$, $\{5, 11\}$ and $\{6, 12\}$ (Figure 6a). Each fold is predicted by a model trained on the remaining ten months with a fixed number of boosting iterations. The out-of-fold predictions of each stage serve as the training input of the next stage, so that every stage is validated under the conditions in which it is applied. The RMSE is reported on all records, on all records except the five corrupt records of 2025 whose label exceeds 20,000 s although take-off occurred within 45 min of the schedule, and on flights without recording defects (label within 3,600 s of the base prediction).

5.2 Selection rule

Let $M = \{1, 2, 6, 7\}$ denote the 2025 analogues of the evaluation months and $\Delta_m(w)$ the change in RMSE in month m when a candidate component enters the complete pipeline with weight w, on top of all previously admitted components. With w^* the optimum over all twelve months on a fixed grid, the candidate is admitted only if the conditions of Equation 6 hold,

$$\Delta_m(w^*) < 0 \quad \forall m \in M, \qquad \Delta_M(w^*) < 0, \qquad G_{100} = \frac{\sum_{i \in \text{top } 100} g_i}{\sum_i g_i} < 1, \tag{6}$$

where g_i is the reduction in the squared error of record i over the records of M. The concentration criterion $G_{100} < 1$ rejects candidates whose gain rests on a small number of records, for the reason given in Section 3. For a change that requires re-training the whole pipeline, such as a new ADS-B extraction, the records whose inputs are unchanged differ only by re-training noise (about 20 s in root mean square on covered records), and G_{100} is computed over the records whose inputs changed. The weight is additionally selected on five folds and evaluated on the sixth, which removes the optimistic bias of selecting and assessing on the same data [11]. Before an evaluation file is produced, the root-mean-square change that a component induces on 2026 is compared with the corresponding change on 2025, and a ratio above 1.5 is treated as a failure. The admissible scopes of a component are fixed before results are observed, and no leaderboard score enters any fit, weight or threshold; every decision is stored in a file. One decision was taken after a leaderboard score was known: the arrival-located stands improved every analogue month but failed $G_{100} < 1$ over all records (2.43); the restriction to the records whose inputs changed was introduced after their January/July leaderboard score had been seen, and under it they pass ($G_{100} = 0.69$ over 2,945 records).

5.3 Audit

Every component admitted by 12 October 2026 was re-evaluated in the configuration of that date by removing it in turn and by selecting its weight in nested cross-validation; the components admitted afterwards (the mixed-target LightGBM member, the assigned-stand timing, stage 4 and the arrival-located stands) were each checked fold by fold at admission. All components reduced the four-month error under nested weights; stage 1b increased the February error by 0.03 s, which is within noise, and reduced the error in the other three months. The nested weights equalled the selected weights in five or six of the six folds, except for the decorrelators, for which 0.075 was preferred to 0.05 in five folds at the same four-month gain. The audit also showed that the parameters of the reconciliation layer had been estimated on all of 2025 in the reported validation; re-estimating them per fold changed the four-month RMSE by 0.24 s and altered no decision.

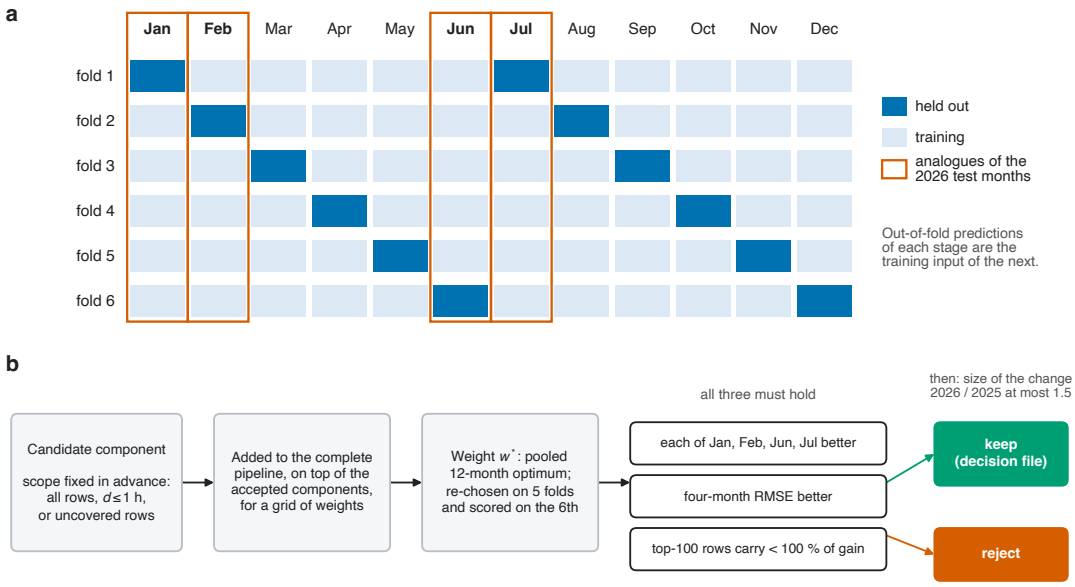


Figure 6. (a) Six two-month folds of 2025; the months framed in red are the analogues of the 2026 evaluation months. (b) Selection rule applied to every candidate component.

6. Results

Table 4 reports the 2025 out-of-fold RMSE of the final pipeline. The all-record value for January is dominated by the five corrupt records. On the organisers' public leaderboard, which scores January and July 2026, the January and July predictions of the submitted file attain an RMSE of 234.55 s, close to the 237.9 s of the 2025 January and July analogue without the corrupt records; the final ranking over the four months is published by the organisers. Table 5 traces the first stages of the pipeline from a baseline that predicts the NM clock anchor alone (447.0 s over the four months), which the base learners reduce to 273.6 s, Table 6 the components admitted subsequently, and Figure 7 the concentration of the error and the contribution of each admitted component. The ADS-B stacker yields the largest reduction (13.7 s over the four months), followed by the daily reconciliation (8.7 s, with every month improved). The two mixed-target members were admitted, whereas a deeper CatBoost model with the standard residual target was rejected in every scope, indicating that the useful diversity originates from the target formulation rather than from model capacity.

Table 4. 2025 out-of-fold RMSE (s) of the final pipeline on all records, on all records except the five corrupt records, and on records without a recording defect. Each 2025 month is the analogue of the corresponding evaluation month.

2025 month	All records	Without corrupt	No defect
January	324.5	196.7	175.5
February	212.3	212.3	193.5
June	285.7	285.7	186.6
July	266.4	266.4	220.1
Four months	276.0	246.9	195.8
Twelve months	228.9	208.1	179.5

Table 5. First stages of the pipeline: 2025 out-of-fold RMSE (s) per analogue month and over the four months (Four), excluding the five corrupt records; the last column restricts the four months to records without a recording defect.

Model	Jan	Feb	Jun	Jul	Four	No defect
NM clock anchor alone ($\hat{y} = b$)	385.71	411.92	449.10	511.80	447.03	410.73
LightGBM, learning rate 0.05	225.95	255.42	311.56	305.93	280.59	225.07
LightGBM, mean of 0.05 and 0.02	225.08	254.26	311.09	304.31	279.56	223.91
CatBoost	227.64	246.02	310.41	293.31	274.87	224.01
Base combination	223.58	246.13	307.62	294.29	273.57	220.44
+ ADS-B stacker	205.53	232.22	295.11	281.61	259.83	203.17
+ stage 2	205.27	231.28	294.62	280.63	259.14	202.30

Table 6. Components admitted after the first stages: change of the 2025 out-of-fold RMSE (s, excluding the five corrupt records) at the selected weight, evaluated on top of the previously admitted components.

Component	Weight	Jan	Feb	Jun	Jul	Four	Decision
Decorrelators (records with NM clock)	0.05	−0.54	−0.07	−0.19	−0.48	−0.32	admitted
Mixed-target CatBoost, $d \leq 1$ h	0.5	−0.34	−1.32	−0.19	−1.06	−0.69	admitted
Mixed-target CatBoost, all records	0.4	−0.59	−4.06	−1.59	−3.79	−2.52	rejected
Mixed-target LightGBM, $d \leq 1$ h	0.2	−0.26	−0.27	−0.14	−0.08	−0.16	admitted
Assigned-stand timing	–	−1.19	−0.73	−1.31	−0.67	−0.97	admitted
Arrival-located stands	–	−0.28	−0.15	−0.04	−0.08	−0.12	admitted
Stage 1b	1.0	−1.36	−0.36	−0.38	−0.77	−0.67	admitted
Stage 2b	0.75	−0.37	−0.96	−0.11	−0.45	−0.42	admitted
Daily reconciliation	0.75	−3.71	−15.05	−6.79	−10.10	−8.72	admitted
Daily, separate class for stand-timed records	0.75	−0.01	+0.00	+0.16	−0.02	+0.05	rejected
Stage 4: stand readout	per bin	−0.14	−0.08	−0.08	−0.05	−0.08	admitted
Bounds	–	−0.12	−0.02	−0.07	−0.46	−0.19	admitted
Partner without early-day labels	0.25	+0.02	+0.07	−0.11	−0.04	−0.03	rejected

The all-record variant of the mixed-target CatBoost member was rejected because the 100 most-improved records carried 116 % of its gain ($G_{100} = 1.16$).

7. Rejected alternatives

The following alternatives were evaluated on held-out months of 2025 and rejected. They are reported because several of them are natural first choices.

- Removing or down-weighting defective labels, a Huber loss, or a model trained only on records without defects: lower error on records without defects, but higher overall RMSE.
- Rules specific to LIRF, including two-part models of schedule-copied records, regime mixtures and

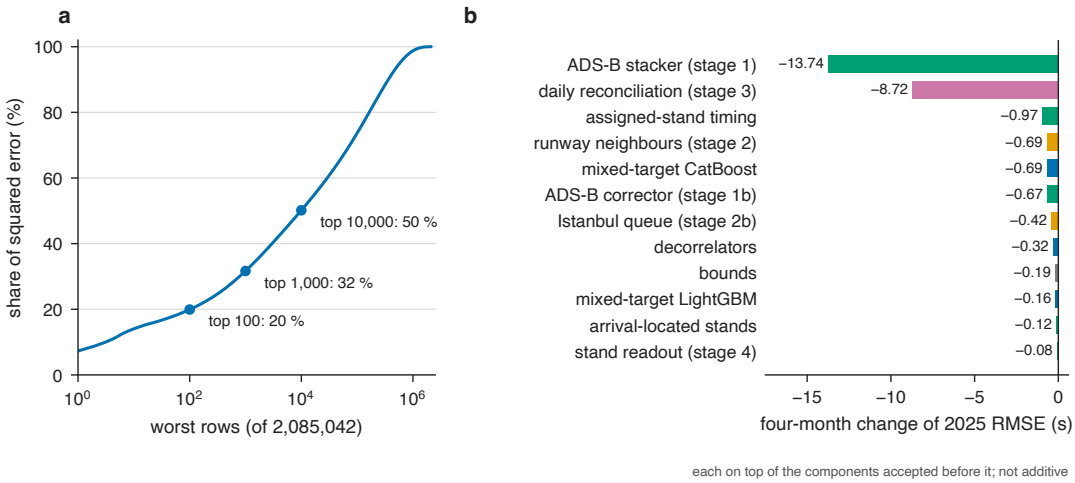


Figure 7. (a) Cumulative share of the 2025 out-of-fold squared error carried by the records with the largest errors in the final pipeline. (b) Change of the four-month 2025 RMSE produced by each admitted component, as in Tables 5 and 6 (each on top of the components admitted before it; the values are not additive).

day-shift hedges: every gain originated from fewer than 100 records.

- A tabular neural network (a BatchEnsemble multilayer perceptron in the style of TabM [12]) combined on uncovered records: improvements in January, June and July, deterioration in February. AutoGluon on a sample of 400,000 records and XGBoost as a third family of tree ensembles yielded no gain.
- A deeper CatBoost model (depth 10) with the residual target, and a member with a multiplicative target (y/b with weight b^2), failed in every scope; the latter increased the July error by 2.0 s at a weight of 0.1, because an error in the ratio is multiplied by a long clock gap.
- Monthly level calibration towards EUROCONTROL's published mean taxi-out times, including an offset modelled by airport, de-icing share and share of records without a reference time: February or June deteriorated at most settings, because the offset with respect to the competition labels varies with the season.
- EUROCONTROL's daily pre-departure delay attributed to air traffic control as a second constraint: its correlation with any daily sum of off-block delays does not exceed 0.54.
- Recovering an NM flight for unmatched departures from their arrival record and moving the prediction outside the join window: July deteriorated, and a handful of records carried most of the gain.
- Constant bias corrections by lane, airport, delay band, clock pattern and hour: the best variant changed the RMSE by +0.01 s.
- Cross-seasonal validation (training on January and testing on July, and vice versa) did not predict which components would persist better than the six-fold protocol.

To determine whether the features retain usable information, a LightGBM model was trained to predict the remaining error of the final pipeline from all features, with the same six folds and restricted to records without defects. Applied to the held-out months, it increased the error in every lane and at every strength (by 0.24 s on records without defects at a strength of 0.25); its most frequently used inputs were the stand, the combination of stand and runway, and the destination, which indicates

that it fitted noise. The available features are therefore exhausted.

8. Discussion

Information content of the data. Covered flights without recording defects reach an RMSE of 155 s, whereas uncovered flights reach 218 s; the difference quantifies the value of observing the actual pushback. The uncovered departures (all of LTFM, and LFPG and LIRF in part) lack open ground surveillance, and the defective records can be identified but their error cannot be sized. The remaining error is therefore dominated by information absent from the data, in agreement with the residual-learnability test. Further progress requires new observations of the off-block moment rather than further modelling of the existing columns. One qualification concerns form rather than content: a tree ensemble represents a precise clock only coarsely, and the explicit readout of stage 4 (Section 4.7) recovers part of the precision of the assigned-stand clock that the stacker leaves unused.

Model selection under a temporal shift. The requirement that each analogue month improve and the concentration criterion are deliberately conservative: they rejected candidates with large 2025 gains, such as the all-record mixed-target member (2.5 s), because gains carried by a few defective records are not expected to recur in another year. This conservatism is considered the most transferable element of the method, since the records that determine a few seconds of RMSE differ from one year to the next. The organisers of the preceding challenge reached a consistent conclusion: team rankings changed markedly between its two final evaluation months, indicating that generalisation across time periods was a principal difficulty [13].

Limitations. The receiver network of adsb.lol changed between 2025 and 2026: in the analogue months the covered share rose from 52–61 % to 65–72 %, and a few receiver-dependent columns remain in the stacker because removing them increased the 2025 error. The reconciliation layer requires the published daily total for the evaluation period; where it is missing, the airport-day is left unchanged.

9. Reproducibility and reuse

A single script runs weather and ADS-B retrieval, feature construction, every learner and every selection step in the order in which they were taken, and writes the evaluation file; each selection step re-derives its decision from 2025, and the decisions underlying the submitted file are stored in the repository. All period-dependent choices are configuration settings (evaluation file, training files, excluded months and analogue months), so that the same code predicts another period or performs a dry run on a held-out month of 2025. On a laptop with 16 GB of memory and eight threads, the processor-bound steps take approximately one day, dominated by the six-fold out-of-fold training of the base learners; the mixed-target CatBoost member requires about 25 min per fold and 25 to 30 min per final seed on two NVIDIA T4 graphics processors and can be omitted at a measured cost of approximately 0.6 s. The feature construction was executed from a fresh copy of the repository and reproduced the training table exactly, and the complete script was executed end to end from a fresh copy in a reduced configuration before the last two steps (stage 4 and the arrival-located stands) were added; both were run on the full data to produce the submitted file. Three components can be reused independently: the ADS-B ground-track extractor, the reconciliation of predictions with a published daily total, and the selection rule.

10. Conclusion

The prediction of taxi-out times at the airports considered is primarily the estimation of an unobserved moment from imperfect clocks. Treating the aircraft's own ground track as an additional clock, the NM join rule as a hard bound and a published daily delay total as a joint constraint produced the largest improvements, while gradient-boosted residual learners with target-diverse members and neighbourhood corrections provided the remainder. A single conservative selection rule kept out gains carried by a few records. The remaining error is concentrated where no open data observe the pushback and in recording defects whose magnitude no available field predicts.

Appendix 1. Model settings

Table 7 lists the settings of every model. All LightGBM models minimise the squared error with bagging at every iteration, and the numbers of boosting iterations are fixed (no early stopping on held-out months). The CatBoost models use the squared error, a one-hot encoding for categorical features with at most eight levels and target statistics otherwise. Final predictions average the listed number of random seeds.

Table 7. Settings of the models. Rate: learning rate; Iter.: boosting iterations; Leaves: maximum number of leaves (CatBoost: tree depth); Min. leaf: minimum number of records per leaf; Col./row: feature and bagging fractions; L2: L2 regularisation of leaf values.

Model	Rate	Iter.	Leaves	Min. leaf	Col./row	L2	Seeds
LightGBM base, rate 0.05	0.05	760	511	100	0.7/0.8	50	4
LightGBM base, rate 0.02	0.02	1,650	511	100	0.7/0.8	50	3
CatBoost base	0.08	1,500	depth 8	–	–	–	3
LightGBM decorrelators (4)	0.05	760	511	100	0.7/0.8	50	1
CatBoost, mixed target	0.06	3,000	depth 10	–	–	–	8
LightGBM, mixed target	0.05	760	511	100	0.7/0.8	50	3
LightGBM stage 1 (stacker)	0.03	800	511	100	0.7/0.8	50	3
LightGBM stage 1b	0.03	800	63	40	0.8/0.8	10	1
LightGBM stage 2	0.03	400	31	500	0.8/0.8	50	1
LightGBM stage 2b	0.05	300	15	2,000	0.9/0.8	10	1

The mixed-target members train two heads, one per delay regime (Equation 2); the CatBoost member's head for departures more than one hour late is limited to depth 8. The LightGBM base models additionally use a categorical smoothing of 200, a categorical L2 regularisation of 10, 255 bins and one-hot encoding for categorical features with at most eight levels.

Appendix 2. Features and their importance

Table 8 groups the 241 features of the base learners and reports the share of the total split gain that each group receives in a LightGBM base model (learning rate 0.05) and in the ADS-B stacker, both fitted on all 2025 records with the settings of Table 7. Table 9 lists the individual features with the largest gain. The complete feature dictionary is part of the repository (`docs/FEATURES.md`). In the base model the clock gaps dominate, which confirms the role of the NM clock as the anchor; in the stacker the ADS-B timings receive the largest share, consistent with the 13.7 s reduction that the stacker provides.

Table 8. Feature groups: number of features in the base learners and share of the split gain (%) in the base model and in the ADS-B stacker. The stacker uses the base features, 36 ADS-B timing features and the base prediction.

Feature group	Features	Base model	Stacker
NM clocks and clock gaps	25	58.4	12.4
Runway queue and traffic	85	12.4	11.9
Target encodings	16	9.3	5.3
Aircraft, operator, stand and runway identity	16	6.8	6.5
Arrivals, stands and references	22	5.4	5.1
Weather and de-icing	38	2.7	2.7
Neighbour NM clocks	17	2.2	2.6
Recording-defect state	10	1.9	1.9
Flight and route	5	0.6	0.4
Calendar	7	0.4	0.5
ADS-B ground-track timing	–	–	41.6
Base prediction	–	–	9.2

Table 9. Features with the largest share of the split gain (%) in the base model (top) and in the ADS-B stacker (bottom).

Feature	Definition	Gain (%)
`naive_raw_minus_unimp`	NM clock gap minus the unimpeded taxi time of the stand and runway	16.5
`aobt3_minus_eobt1`	NM AOBT minus the first estimated off-block time	12.1
`naive_minus_unimp`	plausible NM clock gap minus the unimpeded taxi time	8.3
`aobt3_minus_lobt`	NM AOBT minus the LOBT	4.5
`tko_minus_sched`	delay to take-off, d	2.9
`ter_stand_rwy_mean`	out-of-fold mean residual of the stand and runway	2.5
`ter_apt_op_mean`	out-of-fold mean residual of the airport and operator	2.1
`naive_raw`	NM clock gap, take-off minus NM AOBT	1.7
`base`	base prediction	9.2
`ads_first_slow`	first ground sample at 1 kn or more	4.0
`ads_movsust_minus_aobt`	first sustained movement minus the NM AOBT	3.9
`ads_first_mov_sust`	first movement above 5 kn sustained over two samples	3.7
`ads_first_mov`	first ground sample above 5 kn	3.7
`ads_mov_minus_aobt`	first movement minus the NM AOBT	3.6
`ads_as_dmin`	closest distance of the track to the assigned stand	2.6
`stand`	assigned stand	2.3
`ads_path_m`	length of the observed taxi path	2.1
`ads_first_fast`	first ground sample above 15 kn	2.0

Acknowledgement

The author thanks the adsb.lol contributors, the Iowa Environmental Mesonet, EUROCONTROL and the OpenSky Network [14] for the data, and the developers of LightGBM, CatBoost, pandas, NumPy, scikit-learn and PyArrow. The work also benefited from open-source repositories and write-ups published online.

Open data statement

The movement records were provided by the organisers of the PRC Data Challenge 2026 [3]. The external data are openly available: the adsb.lol daily archives under the Open Database License [4], the Iowa Environmental Mesonet ASOS/METAR archive [5], and EUROCONTROL's all pre-departure delay files [6]. The aggregated data underlying the figures are included in the repository (`docs/paper/figure_data.json`).

Reproducibility statement

The source code is available at https://github.com/piyush7911/prc2026 under the GNU General Public License v3. The script `scripts/run_final.sh` reproduces the evaluation file from the raw data, including every selection decision; the decisions of the submitted file are stored in `docs/decisions`. The scripts `figure_data.py` and `make_figures.py` in `docs/paper` generate the figures, and the values in the tables are written by the pipeline to `validation_2025.json` in `outputs/final` and to the decision files, except the rows of the two changes to the ADS-B extraction, which `prc.validate.compare` produces, and the bounds row (the assembly with the bounds switched off); the baseline and the appendix tables are produced by `feature_importance.py` in `docs/paper`.

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